

Vhilly Manalansang

+63 956 480 8269 | vhillymanalansang02@gmail.com | <https://github.com/yukonseeme>

EDUCATION

Bachelor of Science in Computer Science University of the East Caloocan	August 2022 – April 2026 (Expected) Caloocan, Metro Manila
Senior High School - Information and Communication Technology University of the East Caloocan	August 2020 - 2022 Caloocan, Metro Manila

WORK EXPERIENCE

Backend Developer (On the Job Training) Mr. Geek Philippines	June 2025 – July 2025 Metro Manila
- Developed and maintained RESTful APIs to ensure seamless data communication between the server and client-side applications. - Containerized applications using Docker, streamlining the deployment process and ensuring environment consistency across development and production. - Collaborated with the development team to design scalable backend architecture for ready-to-deploy software solutions.	

PROJECTS

ManDiligma Arduino and React/TypeScript	September 2025 – December 2025
- Developed a Smart Irrigation system by implementing a K-Nearest Neighbors (KNN) machine learning algorithm on Arduino hardware, integrated with a React mobile application. - Trained and optimized the KNN model to automate irrigation logic, successfully syncing real-time sensor data with the database for high-precision water management. - Increased watering efficiency by utilizing predictive modelling to prevent over-irrigation based on soil moisture, temperature and humidity.	
Combined Cryptographic and Steganographic Techniques for Image Metadata Security 	September 2024 – December 2025
- Implemented robust data protection by applying AES (Advanced Encryption Standard) to secure image metadata, preventing unauthorized access to sensitive information. - Engineered a secure user authentication system by implementing Password Hashing process with Argon2 using Python to ensure industry-standard credential storage. - Designed an optimized database architecture with SQLite that functioned seamlessly across different platforms for 91.63% of evaluators. - Validated software quality standards (ISO 25010) through a large-scale evaluation of 215 participants, maintaining high internal consistency with Cronbach’s alpha values up to 0.96.	

TECHNICAL SKILLS

Languages | Java, Python, JavaScript, C++, SQL
Web & Frameworks | React, Django, Angular, Express.js, Firebase
DevOps & Tools | Git, Docker, MySQL, Redis, Arduino
Data Science | NumPy, Pandas, Matplotlib, Bootstrap, TensorFlow